

Reflection Log

Credit Name: CSE3910 - Project D

Assignment Name: Move

How has your program changed from planning to coding to now?

The initial plan for this assignment was to control the rover's forward and backward movement. While straightforward, several improvements were made during development to optimize performance and usability.

1. Over-Acceleration Issue

The rover accelerated too quickly at full speed, making its movement jerky. I reduced the motor velocity for smoother acceleration.

Code Example:

```
leftMotors.setTargetVelocity(0.8); // Reduced speed for smoother movement  
rightMotors.setTargetVelocity(0.8);
```

2. Improved Timing Control

The program originally used hardcoded delays, which didn't always align with the rover's real-time performance. I adjusted the timing dynamically based on testing results.

Code Example:

```
Thread.sleep(1500); // Adjusted for consistent travel distance
```

3. Backward Movement Refinement

The backward movement was less reliable due to uneven motor performance. I added a small offset to equalize motor speeds during reverse.

Code Example:

```
leftMotors.setTargetVelocity(-0.8);  
rightMotors.setTargetVelocity(-0.85); // Slight adjustment for balance
```

4. Resource Cleanup

As with other assignments, I noticed that motor connections weren't closed properly, so I

added cleanup logic.

Code Example:

```
leftMotors.close();  
rightMotors.close();  
|
```

Final Thoughts: The adjustments ensured smoother movements and improved the program's reliability. The rover now transitions between forward and backward motion seamlessly.